

```
# Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.preprocessing import StandardScaler
import plotly.express as px
import plotly.graph_objects as go
```

```
# Load the dataset
df = pd.read_csv("/content/AQI-and-Lat-Long-of-Countries.csv")
```

```
# Display basic information about the dataset
print(df.head())
print(df.info())
```

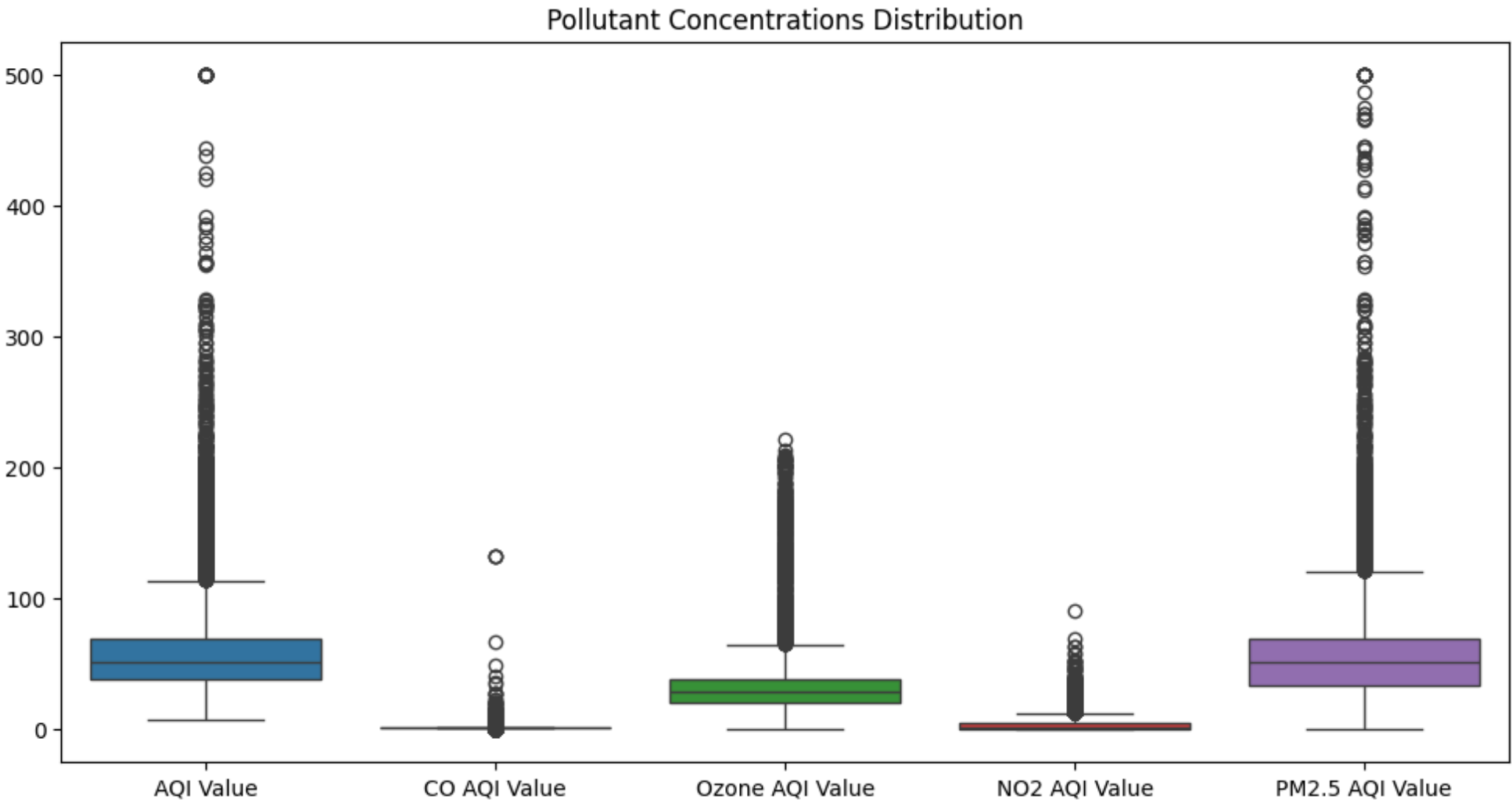
↗

	AQI Value	CO AQI Value	Ozone AQI Value	N02 AQI Value	PM2.5 AQI Value	\
0	51	1	36	0	51	
1	41	1	5	1	41	
2	41	1	5	1	41	
3	66	1	39	2	66	
4	34	1	34	0	20	

	lat	lng
0	44.7444	44.2031
1	-5.2900	-44.4900
2	-11.2958	-41.9869
3	37.1667	15.1833
4	53.0167	20.8833

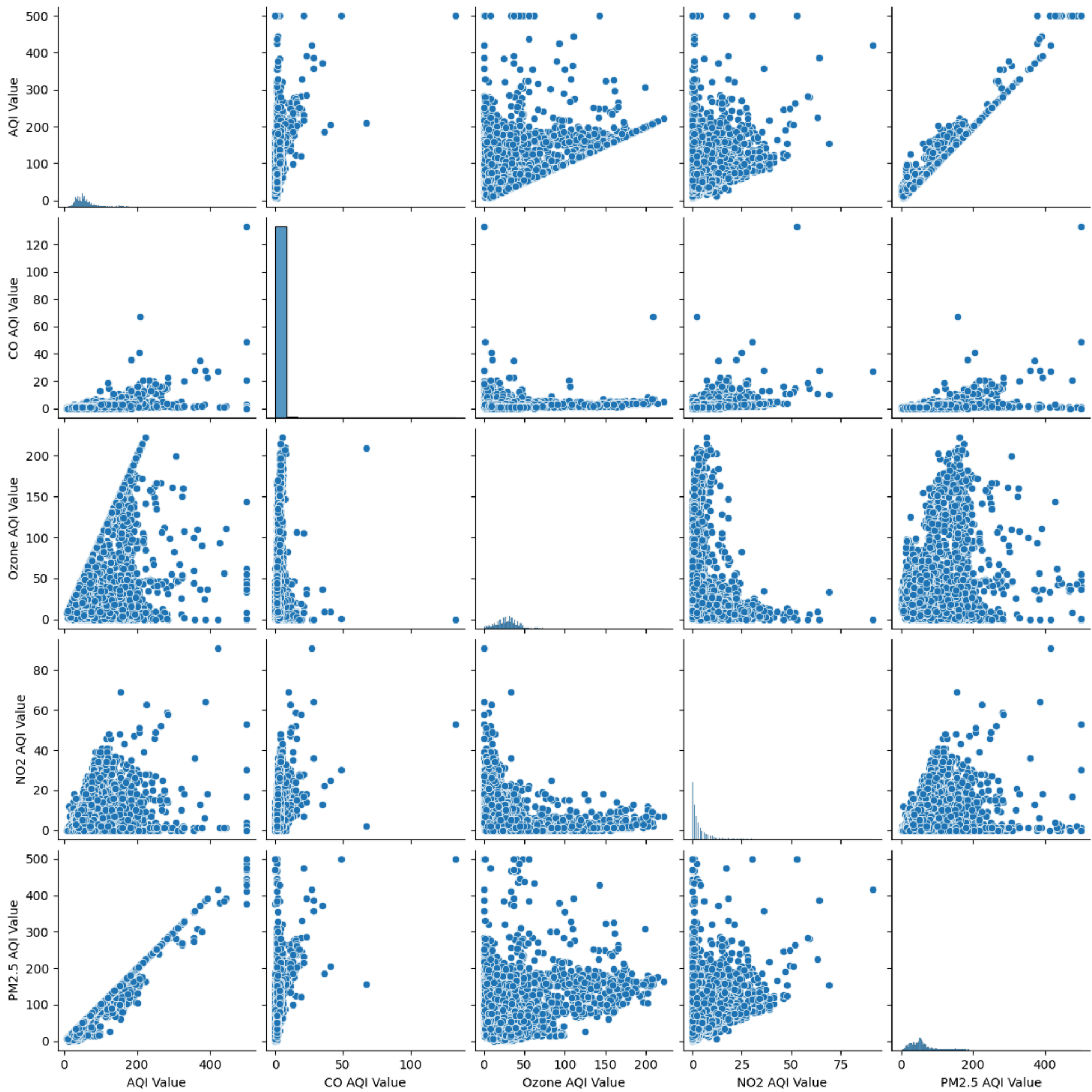
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 16695 entries, 0 to 16694
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   AQI Value              16695 non-null  int64
1   CO AQI Value           16695 non-null  int64
2   Ozone AQI Value        16695 non-null  int64
3   N02 AQI Value          16695 non-null  int64
4   PM2.5 AQI Value        16695 non-null  int64
5   lat                    16695 non-null  float64
6   lng                    16695 non-null  float64
dtypes: float64(2), int64(5)
memory usage: 913.1 KB
None
```

```
# Exploratory Data Analysis (EDA)
# Plotting pollutant concentrations
plt.figure(figsize=(12, 6))
sns.boxplot(data=df[['AQI Value', 'CO AQI Value', 'Ozone AQI Value', 'N02 AQI Value', 'PM2.5 AQI Value']])
plt.title('Pollutant Concentrations Distribution')
plt.show()
```

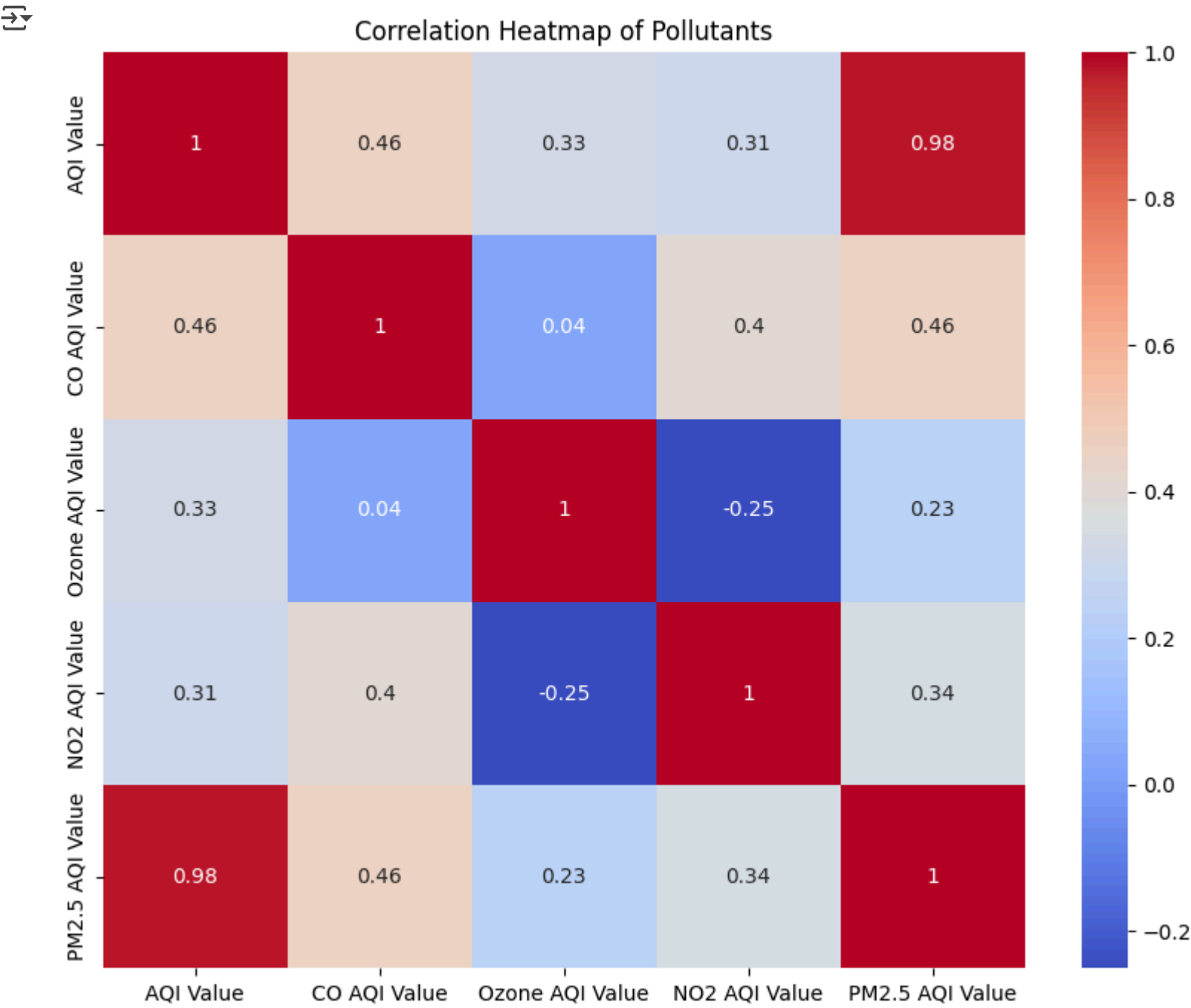


```
# Visualizing the relationship between pollutants
plt.figure(figsize=(14, 7))
sns.pairplot(df[['AQI Value', 'CO AQI Value', 'Ozone AQI Value', 'NO2 AQI Value', 'PM2.5 AQI Value']])
plt.show()
```

<Figure size 1400x700 with 0 Axes>



```
# Correlation heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(df[['AQI Value', 'CO AQI Value', 'Ozone AQI Value', 'NO2 AQI Value', 'PM2.5 AQI Value']].corr(), annot=True,
plt.title('Correlation Heatmap of Pollutants')
plt.show()
```



```
# Feature Selection - We'll use all the pollutants to predict AQI Value (as an example)
X = df[['CO AQI Value', 'Ozone AQI Value', 'NO2 AQI Value', 'PM2.5 AQI Value']]
y = df['AQI Value']
```

```
# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
# Standardize the data
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
# Model Building - Linear Regression
lr = LinearRegression()
lr.fit(X_train_scaled, y_train)
```

LinearRegression ⓘ ?

LinearRegression()

```
# Model Evaluation - Linear Regression
y_pred_lr = lr.predict(X_test_scaled)
print("Linear Regression - MAE:", mean_absolute_error(y_test, y_pred_lr))
print("Linear Regression - RMSE:", np.sqrt(mean_squared_error(y_test, y_pred_lr)))
print("Linear Regression - R²:", r2_score(y_test, y_pred_lr))
```

Linear Regression - MAE: 4.1765907360691035
Linear Regression - RMSE: 6.504410690902979
Linear Regression - R²: 0.9744908345378135

```
# Model Building - Random Forest Regressor
rf = RandomForestRegressor(n_estimators=100, random_state=42)
rf.fit(X_train_scaled, y_train)
```

RandomForestRegressor ⓘ ?

RandomForestRegressor(random_state=42)

```
# Model Evaluation - Random Forest
y_pred_rf = rf.predict(X_test_scaled)
print("Random Forest - MAE:", mean_absolute_error(y_test, y_pred_rf))
print("Random Forest - RMSE:", np.sqrt(mean_squared_error(y_test, y_pred_rf)))
print("Random Forest - R²:", r2_score(y_test, y_pred_rf))
```

➡ Random Forest - MAE: 0.08699311171009284
Random Forest - RMSE: 1.5803208301877338
Random Forest - R²: 0.9984941875969242

```
# Visualizing predicted vs actual AQI Value
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred_rf, alpha=0.5, color='g')
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], color='r', linestyle='--')
plt.title('Random Forest Prediction vs Actual AQI Value')
plt.xlabel('Actual AQI Value')
plt.ylabel('Predicted AQI Value')
plt.show()
```

